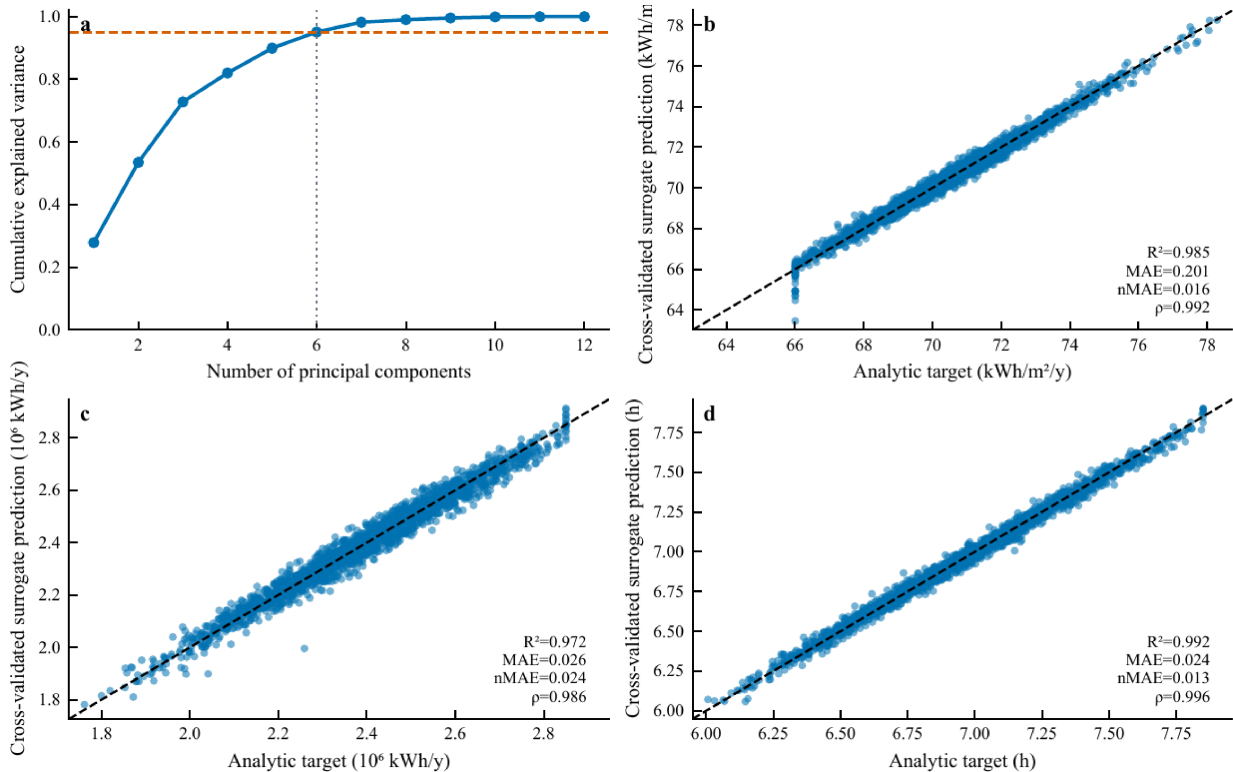




M1 — data_and_surrogate_validation

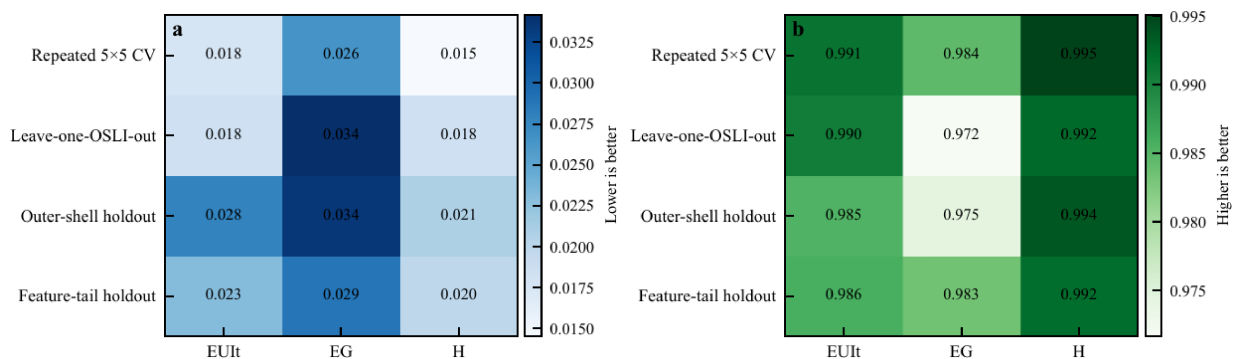
Planned location: Main Fig. 4 candidate

Source files: descriptor_coverage.csv, surrogate_parity_mean_predictions.csv

Source SHA: c9c77a2738ec..., 5837b986d33d...

Claim boundary: Supports descriptor-space coverage and analytic-target surrogate fidelity only.

Unresolved concerns: None



M2 — surrogate_robustness

Planned location: Main Fig. 5 candidate

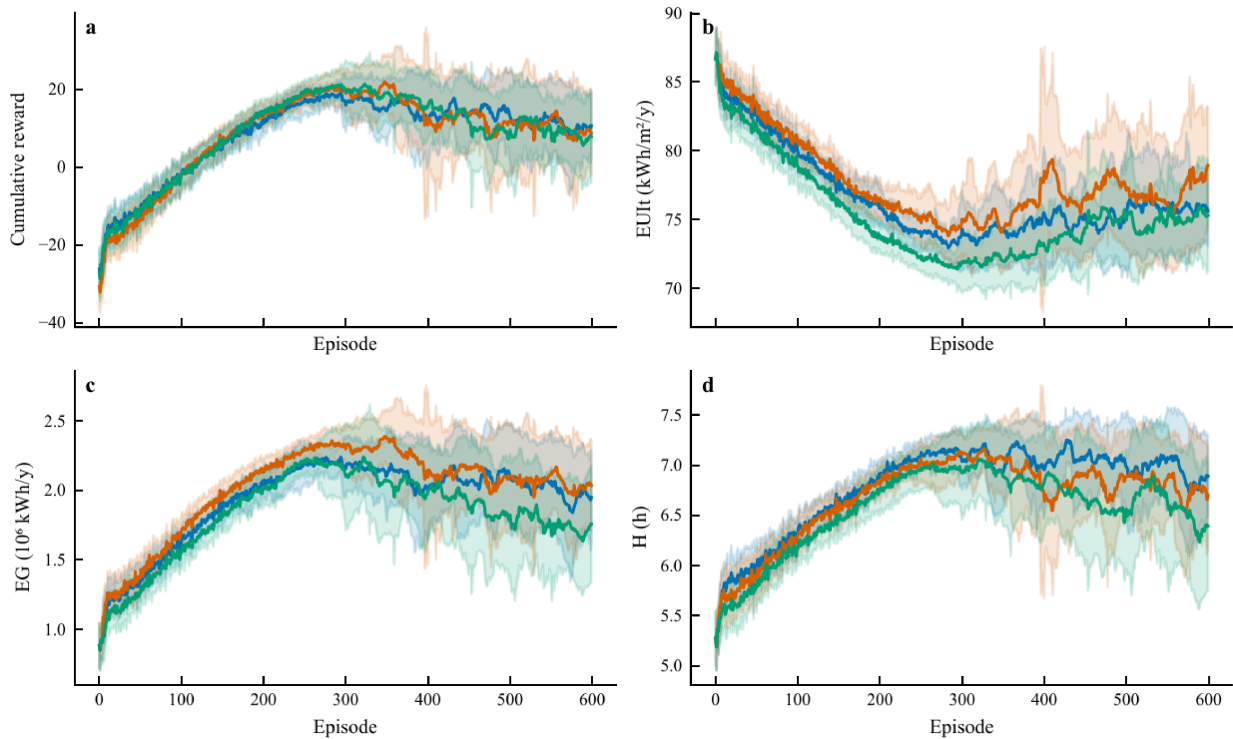
Source files: surrogate_validation_regimes.csv

Source SHA: 4a35f8c39487...

Claim boundary: All panels remain analytic-target surrogate-validation evidence, not physical validation.

Unresolved concerns: None

● Balanced Performance
 ▲ Energy Saving Focus
 ■ Energy Generation Focus



M3 — ddp_g_training_dynamics

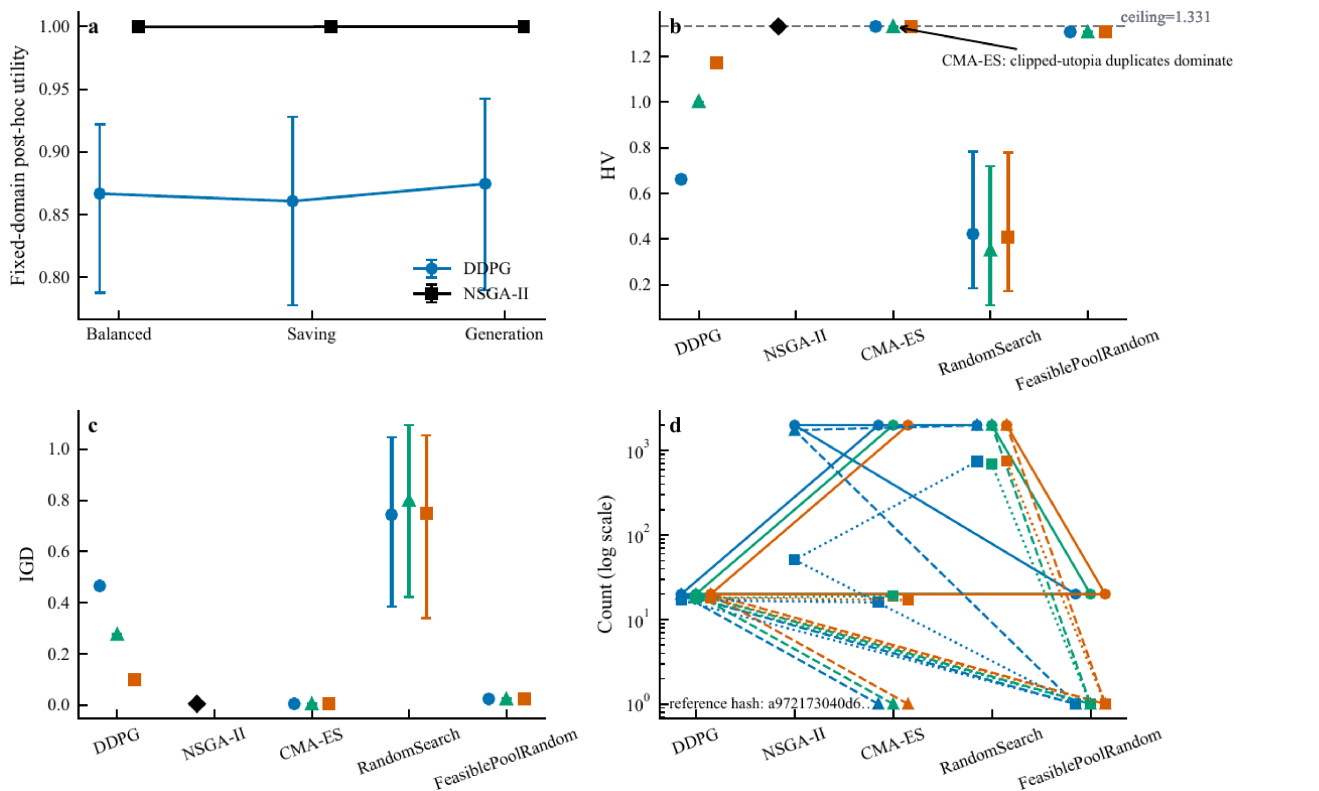
Planned location: Main Fig. 6 candidate

Source files: ddp_g_training_curves_summary.csv, ddp_g_seed_diagnostics.csv

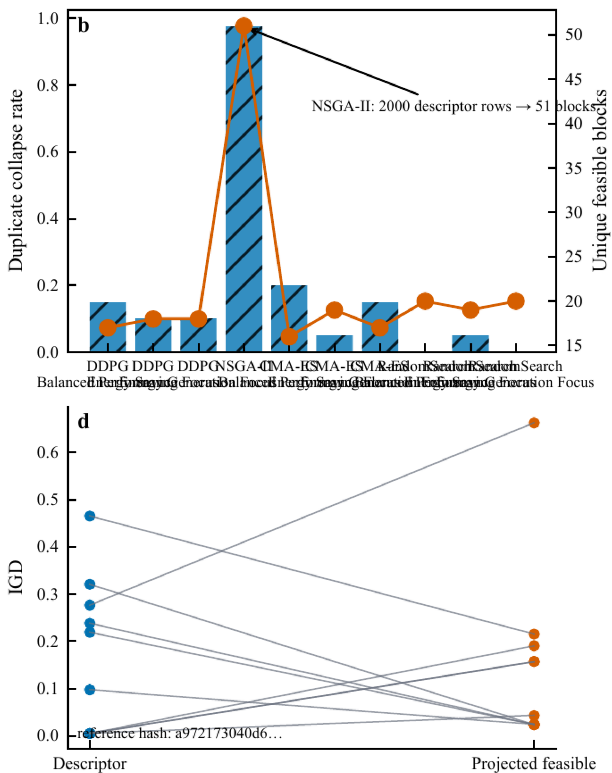
Source SHA: 563e9c7b4738..., ceace85452fb...

Claim boundary: Training dynamics describe serialized surrogate-query search only.

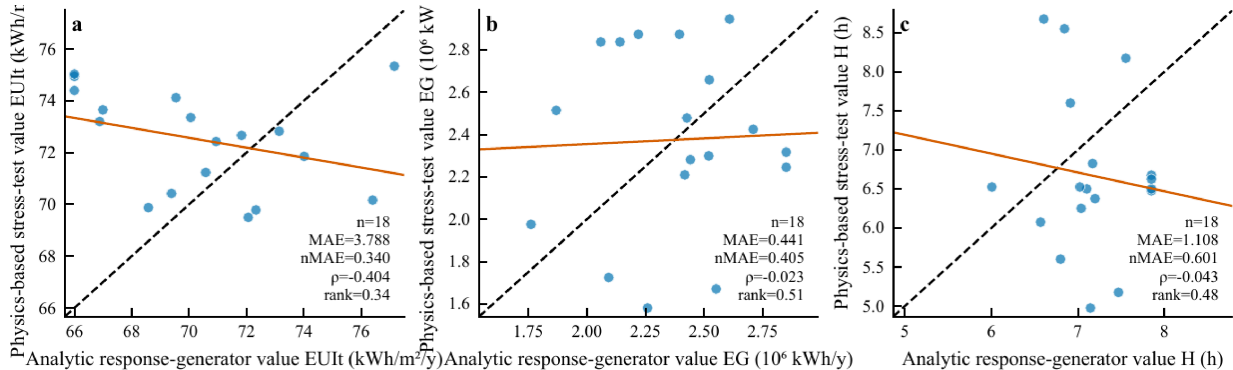
Unresolved concerns: None



M4 — benchmark fairness
 Planned location: Main Fig. 7 candidate
 Source files: benchmark_utility.csv, benchmark_equal_size_20.csv, benchmark_output_contract_counts.csv
 Source SHA: c8dafd40dd57..., f60b07256e7b..., 83c911df9c51...
 Claim boundary: Equal-size metrics are canonical only under benchmark-reference-v2 and must stay separate from asymmetric full-archive diagnostics.
 Unresolved concerns: None



M5 — feasible_projection
Planned location: Main Fig. 8 candidate
Source files: feasible_projection_summary.csv, feasible_projection_metrics.csv
Source SHA: 16f00a0bdf86..., 1132a919d3db...
Claim boundary: Projection panels are representation-sensitivity diagnostics rather than physical validation.
Unresolved concerns: None



M6 — physical_cross_model_stress_test

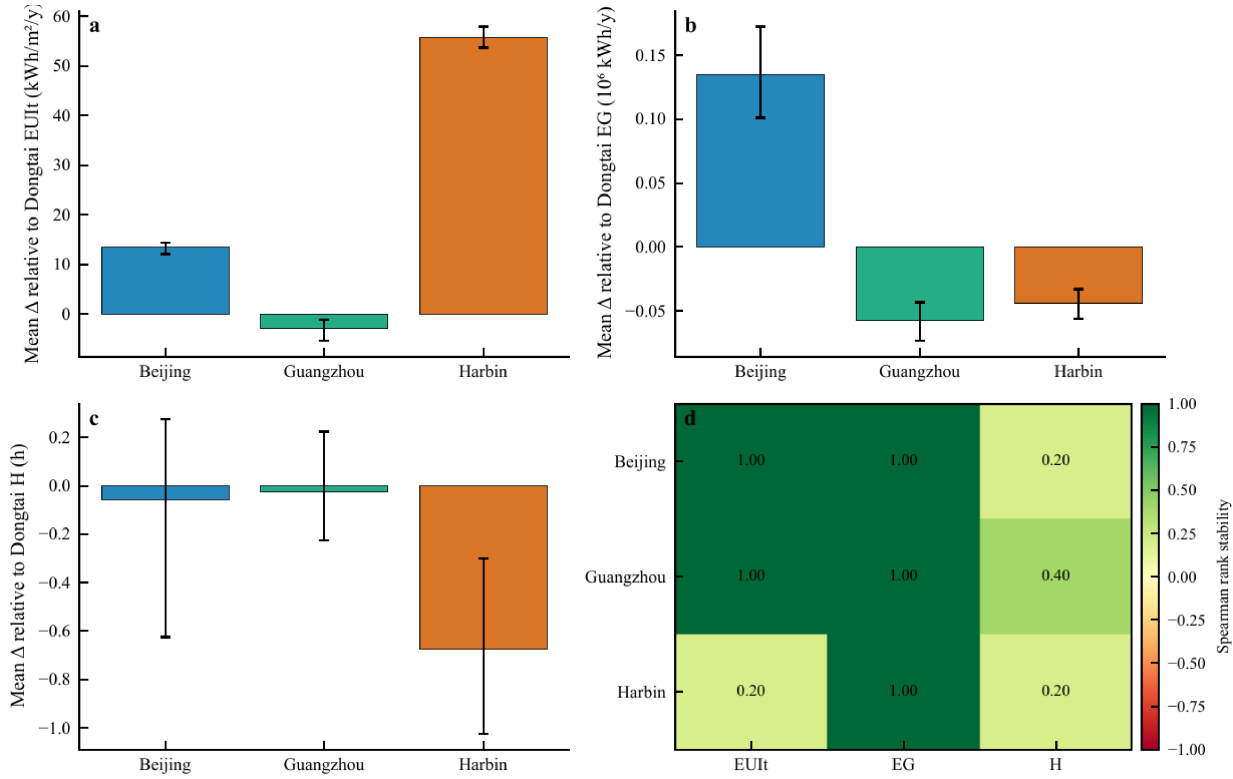
Planned location: Main Fig. 9 candidate

Source files: physical_direct_cases.csv, physical_stress_metrics.csv

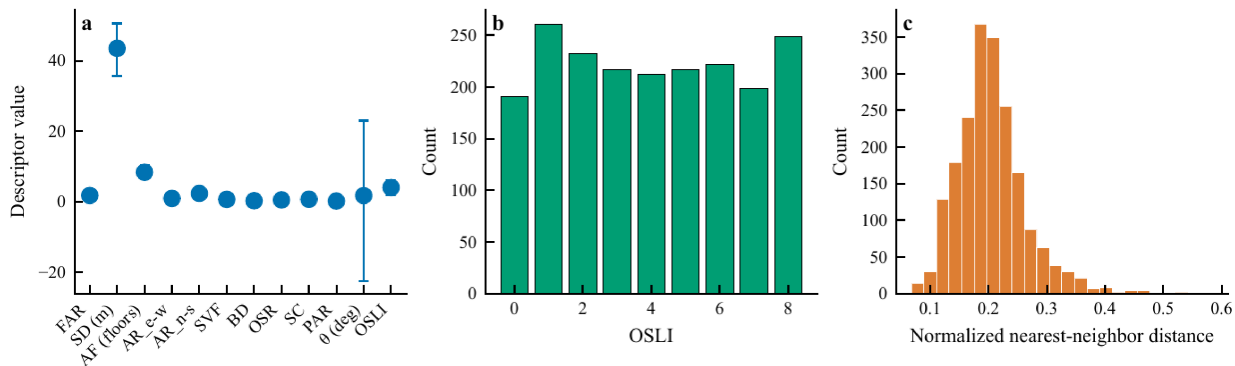
Source SHA: 0403f94ea9bd..., cf9ec8e791d5...

Claim boundary: This figure is limited to the direct-case physics-based cross-model stress test and does not support optimizer-superiority claims.

Unresolved concerns: None



M7 — cross_climate_sensitivity
Planned location: Main Fig. 10 candidate
Source files: climate_case_results.csv, climate_summary.csv, climate_rank_stability.csv
Source SHA: f624253730cd..., 7a03bcf3714f..., ebd3fld2ec54...
Claim boundary: This figure is a limited four-block cross-climate physical sensitivity analysis only.
Unresolved concerns: None



A1 — A1_descriptor_distributions

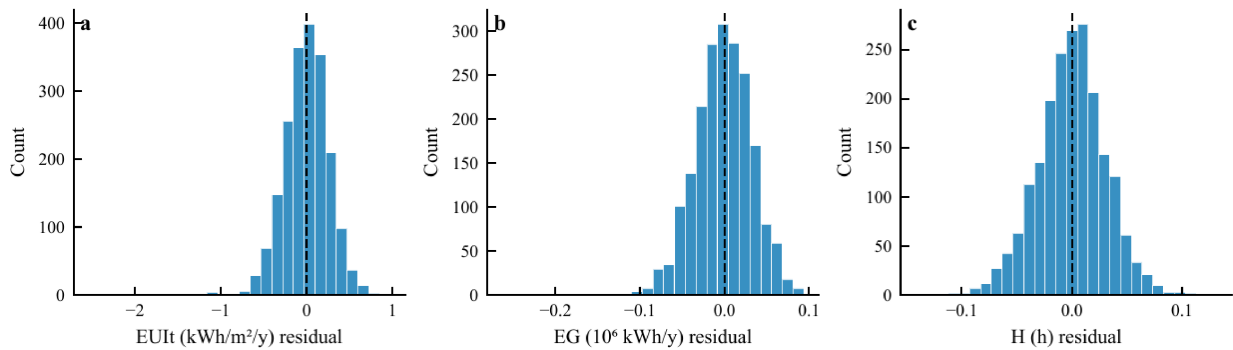
Planned location: Appendix Fig. A1 candidate

Source files: descriptor_coverage.csv

Source SHA: c9c77a2738ec...

Claim boundary: Descriptive coverage diagnostics only.

Unresolved concerns: None



A2 — A2_residual_diagnostics

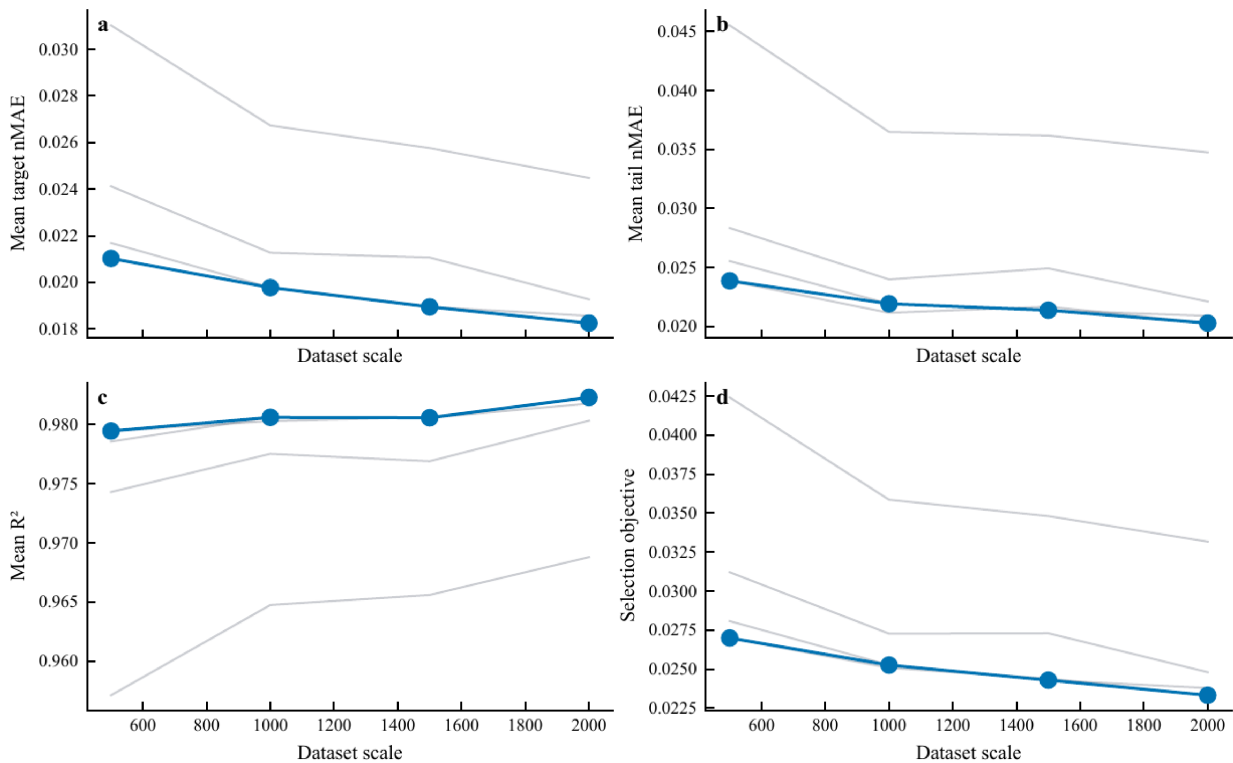
Planned location: Appendix Fig. A2 candidate

Source files: surrogate_parity_mean_predictions.csv

Source SHA: 5837b986d33d...

Claim boundary: Residual diagnostics describe analytic-target surrogate error only.

Unresolved concerns: None



A3 — A3_scale_study

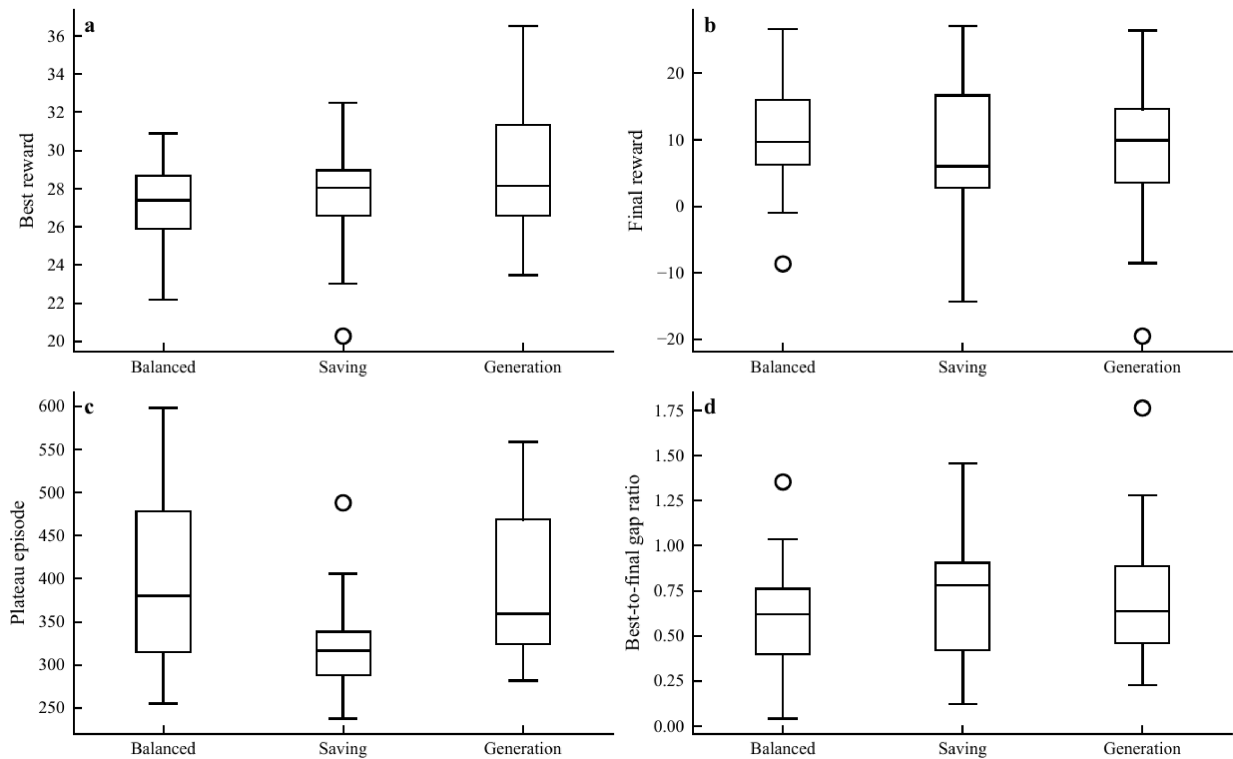
Planned location: Appendix Fig. A3 candidate

Source files: scale_study.csv

Source SHA: 87120cdcbb9...

Claim boundary: Scale-study rows support the surrogate-selection rationale only.

Unresolved concerns: None



B1 — B1_seed_diagnostics

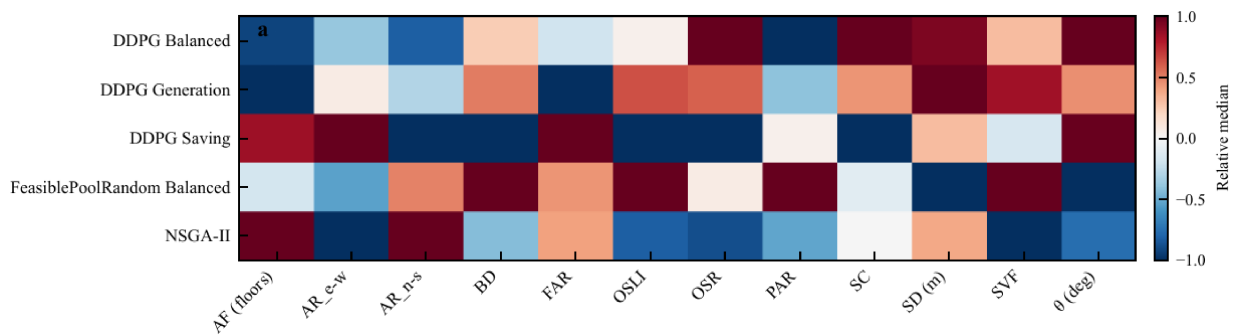
Planned location: Appendix Fig. B1 candidate

Source files: ddpg_seed_diagnostics.csv

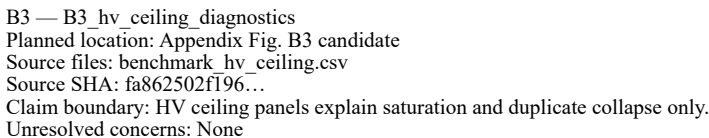
Source SHA: ceace85452fb...

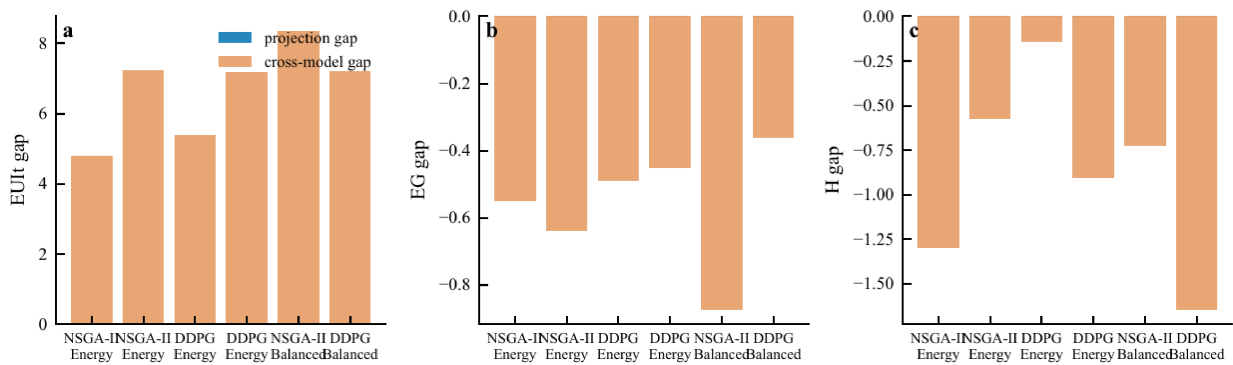
Claim boundary: Seed diagnostics are appendix-only training evidence.

Unresolved concerns: None

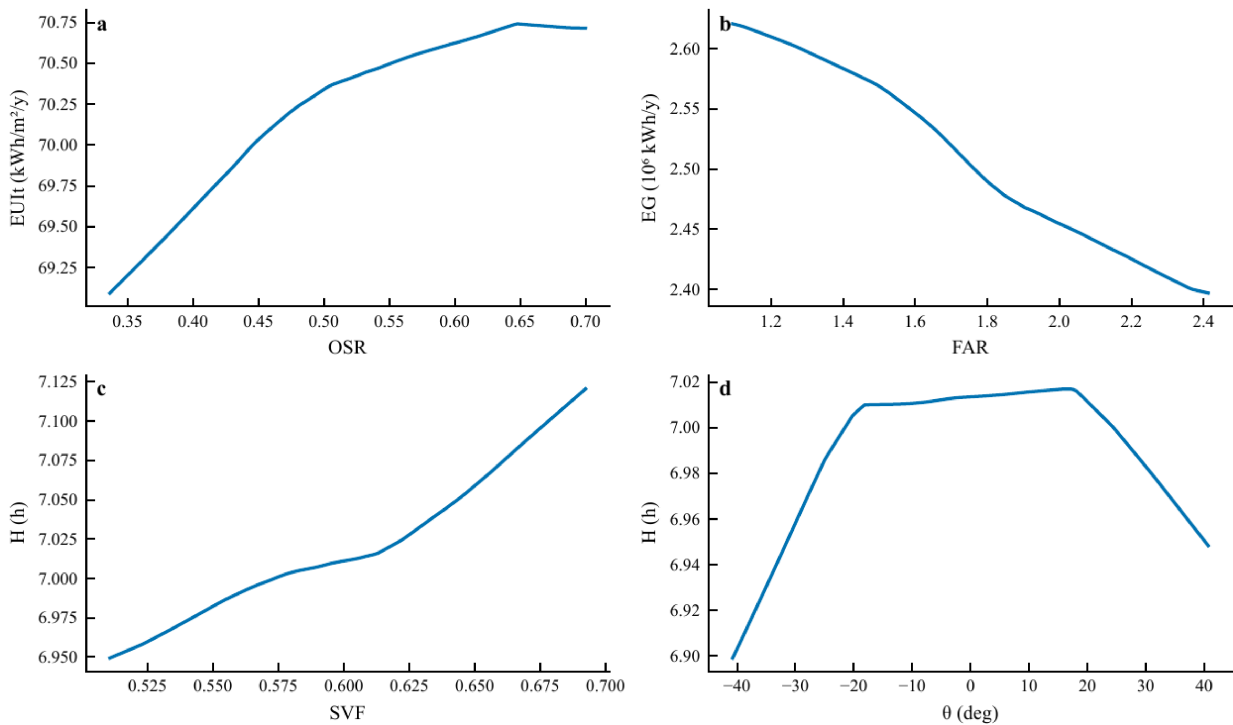


B2 — B2_morphology_signatures
 Planned location: Appendix Fig. B2 candidate
 Source files: morphology_signatures.csv
 Source SHA: aa6df0731235...
 Claim boundary: Descriptor signatures are descriptive summaries, not stable design rules.
 Unresolved concerns: None

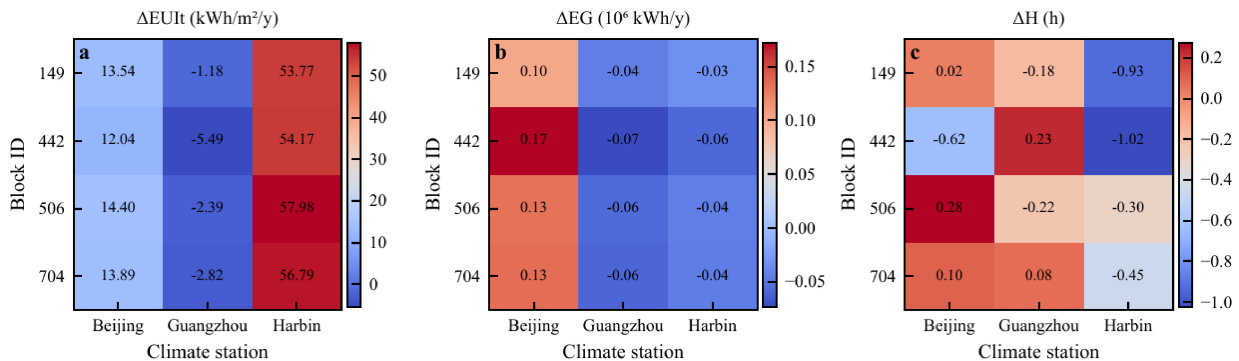




B4 — B4_optimizer_linked_gap_decomposition
Planned location: Appendix Fig. B4 candidate
Source files: optimizer_linked_physical_gaps.csv
Source SHA: 79d6291c8305...
Claim boundary: Optimizer-linked cases remain appendix-only bridge diagnostics.
Unresolved concerns: None



B5 — B5_nonlinear_response_profiles
Planned location: Appendix Fig. B5 candidate
Source files: scale_study.csv
Source SHA: 87120cdcbdb9...
Claim boundary: Selected surrogate response profiles illustrate local trends only.
Unresolved concerns: None



B6 — B6_climate_case_detail
Planned location: Appendix Fig. B6 candidate
Source files: climate_case_results.csv
Source SHA: f624253730cd...
Claim boundary: Case-level climate details remain limited to four blocks and three additional climates.
Unresolved concerns: None